Consensus-Layer Autonomous Code Review: A Layered Multi-Model Monitoring Architecture and a Correlation-Bounded Error Model for Hallucination-Resistant, Human-On-the-Loop Review

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Abstract

Background: Large-language-model coding agents now write a growing share of production code, inverting the classical software economics in which authoring is expensive and review is cheap. Review has become the scarce resource, and the natural response, delegating review to another model, inherits the very failure it must catch: models hallucinate findings, asserting defects that do not exist and, more dangerously, overlooking defects that do. A single automated reviewer cannot be trusted to gate a merge, yet human review capacity does not scale with machine authoring speed.

Aim: We ask whether a structured panel of independent reviewers, running on diverse advanced models and consolidated by agreement, together with recursive monitoring of the consolidator itself, can reduce the residual rate of hallucinated review outcomes far enough to move the human from in-the-loop to on-the-loop (audit-and-exception) for a defined class of changes, and we ask precisely where that reduction stops.

Method: Following a design-oriented approach, we specify a consensus-layer review architecture: N independent review agents on distinct base models, a feedback-reviewer that consolidates by ≥k-of-N corroboration while also verifying panelist conformance, and a meta-monitoring tier that applies the same discipline to the consolidator, generalized to n degrees of monitoring. We formalize per-finding false-positive (hallucination) and false-negative (missed-defect) rates and derive residual rates for the idealized independent case, a shared-latent correlated-error (beta-binomial) case, and the n-tier recursion, backed by an executed analytical and Monte-Carlo computation with two released scripts. The paper's centerpiece is a live experimental protocol (Section 9) that measures the one quantity the model cannot supply about itself, the inter-model error correlation ρ on a labeled corpus of real diffs, with defined estimands, estimators, a power analysis, and pre-registered falsifiable predictions. We specify but deliberately do not execute the live study, on the principle that a simulator which assumes a correlation can only return it.

Results: Under independence, residual hallucination falls exponentially in panel size (from 1.0×10^{-1} at N=3 to 2.3×10^{-10} at N=51 for per-model rate q=0.20 and a 60% corroboration threshold). Under realistic error correlation it does not: the residual is floored at $P(\theta\geq\alpha)$, a quantity independent of both panel size and the number of monitoring tiers, 1.4×10^{-4}, 2.0×10^{-2}, 7.6×10^{-2}, and 1.3×10^{-1} at intra-panel correlation ρ = 0.05, 0.15, 0.30, 0.50. Stacking four monitoring tiers reaches 1.7×10^{-4} only if the tiers are independent; with a 10% common-mode component across tiers the four-tier residual floors near 1.2×10^{-2}. Monte-Carlo (2×10^{6} trials) matches the closed form to within 0.00%.

Conclusion: The naive claim that layered monitoring drives hallucination to zero holds only under a full-independence assumption that models sharing training data and failure modes violate. The honest result is a conditional bound and a correlation floor: consensus review is a powerful reduction, not an elimination, and its ceiling is set by how correlated the panelists' errors are, not by how many reviewers or tiers are added. This is enough to justify human-on-the-loop review with sampled audit for many change classes, but not to remove human sign-off from regulated or irreversible changes. All quantitative results here are analytical or simulated;

the correlation-measurement study of Section 9, which decides where real panels fall on this curve, is the specified and necessary next step.

Keywords: autonomous code review, multi-model consensus, LLM-as-a-judge, hallucination, ensemble error correlation, software supply chain, provenance, regulatory gates.

1. Introduction

Between 2024 and 2026, coding agents moved from autocomplete into authorship: they open pull requests, refactor across files, and land changes with limited human mediation. This inverts a cost structure that software engineering practice has assumed for decades. When writing code was the expensive step, review could be a modest tax on it; a reviewer read a change roughly as fast as a human could produce it. When a model writes the change in seconds, review becomes the binding constraint, and an organization that cannot review at machine speed is not accelerated by its agents, it is accumulating unreviewed risk faster.

The obvious remedy is to review with a model as well. But a model reviewer is subject to the same defect that makes review necessary in the first place. It hallucinates: it reports findings that describe no real defect, and, harder to observe, it misses defects that are present, sometimes confidently signing off. Recent large-scale evaluation of LLM-as-a-judge systems documents that even production judges exhibit position bias, self-enhancement bias, and agreement statistics that collapse once corrected for chance, so a single automated reviewer is not a trustworthy gate for a merge.

This paper studies a specific structure for making automated review trustworthy enough to change the human's role. Rather than one reviewer, a panel of N reviewers runs independently on distinct advanced models and emits structured findings without cross-talk. A feedback-reviewer model consolidates the panel by agreement, findings corroborated across independent reviewers are promoted, and idiosyncratic single-model findings are treated as candidate hallucinations, while separately checking that each panelist produced a well-formed, on-task review. Because the consolidator is itself a model that can fail, a further monitoring tier applies the same discipline to the consolidator, and the construction generalizes to n degrees of monitoring. The intuition offered by practitioners is that with enough independent degrees of monitoring the probability of an uncaught hallucination becomes negligible, approaching a hands-free autonomous reviewer that removes the human from the loop.

We take that intuition seriously and subject it to a formal error model, and our central contribution is to show precisely where it is right and where it fails. Under an idealized independence assumption the intuition is correct and strong: residual hallucination falls exponentially in the number of reviewers. Under the correlation that real models exhibit, shared pre-training corpora, shared tokenizers, shared architectural priors, and the empirically documented tendency to hallucinate the *same* artifacts (for example, the same non-existent package names) repeatedly, the residual does not fall to zero. It is floored by a quantity that depends on the error correlation and is independent of both the panel size and the number of monitoring tiers. The honest headline is therefore a conditional bound with a correlation floor, not an elimination.

We position the contribution against three literatures rather than claiming novelty in any single one. Consensus over sampled model outputs is the premise of self-consistency [1] and multi-agent debate [2]; LLM-as-a-judge and its reliability limits are actively studied [3,4]; and ensemble theory has long held that error *correlation*, not member count, bounds ensemble gains [5]. Our specific contributions are (1) a layered architecture in which the object monitored is the review process itself, recursively, including monitoring of the monitor; (2) a correlation-aware error model that yields the floor result and the diminishing-returns behavior explicitly; (3) the integration of this model with the accountability, regulatory-gate, and software-supply-chain-security constraints that determine whether autonomous review may responsibly replace human review, and where it may not; and, as the paper's centerpiece, (4) a fully specified live experimental protocol (Section 9) for measuring the inter-model error correlation on which the entire architecture's viability turns, with defined estimands, estimators, a power analysis, and pre-registered falsifiable predictions. The first three contributions establish that the whole question reduces to one measurable number; the fourth specifies how to measure it.

Demarcation. We claim design and integration for the architecture as a whole, and empirical support only where an experiment backs it. Every number reported here comes from an executed analytical and Monte-Carlo computation over the error model, not from live model panels. Whether real, diverse models exhibit low enough error correlation for the bound to be useful in practice is the one question the model cannot answer about itself; it requires live multi-model runs, which we specify (Sec. 9) and exclude from execution on principle, exactly because a simulator that assumes an independence level would merely confirm the independence it was told to assume.

2. Background and Definitions

Coding agent. An LLM-driven system that produces a code change (a diff) in response to a task.

Review agent (panelist). A model that reads a diff and emits a set of structured findings, each naming a location, a class (e.g., logic error, security defect, non-existent dependency), and a severity. Panelists run on distinct base models and do not observe one another's findings.

Feedback-reviewer (consolidator). A model that ingests all panel findings and produces the consolidated review by corroboration, promoting a finding only when at least k of the N panelists independently report a matching finding, and additionally performing a conformance check that each panelist produced a valid, on-task, non-degenerate review.

Meta-monitor. A model, or set of models, that applies the consolidator's own discipline to the consolidator's output, detecting consolidator failure modes. Stacking meta-monitors yields n degrees of monitoring.

Hallucinated finding (false positive). A promoted finding that corresponds to no real defect. **Missed finding (false negative).** A real defect present in the diff that the consolidated review fails to report. The two are not symmetric in cost: a false positive wastes reviewer attention, while a false negative ships a defect, and a *shared* false negative, one that every correlated panelist misses, is the architecture's most dangerous case.

Independence and correlation. Two panelists' errors are independent if the event that one errs on an item carries no information about whether the other errs. Correlation ρ measures the degree to which errors co-occur across panelists on the same item, arising from shared data, shared training objectives, and shared failure modes.

3. Related Work

Consensus over model samples. Self-consistency [1] samples multiple reasoning chains from one model and takes a majority vote, improving accuracy on reasoning tasks; multi-agent debate [2] has several model instances propose and critique answers over rounds, improving factuality and reducing hallucination. Both establish that agreement across samples raises confidence, and both are premised, implicitly, on the samples not sharing the same error. Our architecture differs in using deliberately heterogeneous base models rather than samples from one, in consolidating over structured findings rather than a single answer, and in monitoring the consolidation process recursively.

LLM-as-a-judge and its limits. Using a model to evaluate model output is now standard [3], but systematic evaluation reports serious reliability problems: universal chance-corrected agreement deflation, judge rankings that shift by many positions across benchmarks, and strong position and self-enhancement biases even in production judges [4]. These results are the reason a single automated reviewer is insufficient and motivate corroboration across independent judges and explicit monitoring of the judge.

Ensemble theory and error correlation. Classical ensemble learning holds that the benefit of combining classifiers is governed by the diversity of their errors, not merely their number: an ensemble of three identical strong classifiers can be worse than three weaker but decorrelated ones [5]. This is the theoretical backbone of our floor result, adding reviewers helps only to the extent their errors are uncorrelated, transplanted from label classification to the setting of hallucinated review findings across large models that share training data.

Software-supply-chain and provenance controls. The provenance and integrity of machine-authored code is governed by an emerging stack: build-provenance and integrity levels (SLSA), software bills of materials (SBOM), static and dynamic analysis (SAST/DAST), and policy-as-code admission control (OPA/Kyverno) [7,8]. Package-hallucination research [6] quantifies a supply-chain vector directly created by code-generating models. We treat these not as competitors to consensus review but as the deterministic gates that consensus review feeds and must not be permitted to bypass.

Closest prior art, and what this paper does not claim. Each mechanism this paper composes has direct precedent, and we claim neither the panel, the consolidator, nor recursive monitoring as new. Heterogeneous multi-model consensus with a dedicated consolidation model is the design of Council Mode, which reports a 41.7% relative reduction in hallucination on a held-out set [9], and multi-LLM consensus code review already exists as practitioner tooling. Recursive and nested oversight, an overseer monitoring an overseer, is an established line in scalable oversight, including recursive self-critiquing [11], scaling laws for oversight [12], and weak-to-strong monitoring of agents [13]. The intrinsic-limit result, that hallucination cannot be eliminated by architecture or data alone, is argued directly by fundamental-limits work [10], and generalized ensemble bounds that account for sample correlation have been derived [5]. Our contribution is therefore an integration rather than a new primitive: the specific synthesis of a diverse review panel, a conformance-checking consolidator, and n-tier monitoring, cast as a single correlation-bounded error model whose floor is invariant to both panel size and tier count, and tied to a governance rule (the floor sets where the human may move on-the-loop and where human sign-off must remain) and to the software-supply-chain and accountability gates. Where prior work reports an empirical reduction on one configuration, we characterize the reduction's ceiling and its dependence on a single measurable quantity, inter-model error correlation, and we are explicit that this quantity is assumed here and must be measured live.

4. The Consensus Review Architecture

4.1 The independent review panel

The panel is N review agents on distinct base models, each receiving the same diff and task specification and emitting structured findings independently. Independence is the load-bearing property of the whole design, and it is not free: it must be engineered along several axes, **model diversity** (different providers and base models, not different prompts to one model), **prompt diversity** (varied rubrics and framings so panelists do not share a single blind spot introduced by prompt wording), **sampling diversity** (independent seeds and temperatures), and **context partitioning** (no panelist sees another's output). The error model of Section 6 makes explicit why each axis matters: every shared component raises ρ and therefore raises the floor.

4.2 The feedback-reviewer: consolidation and conformance

The consolidator plays two roles. First, **consolidation by corroboration:** it clusters semantically matching findings across panelists and promotes a finding only when at least k of N panelists report it, so that an idiosyncratic single-model assertion is treated as a candidate hallucination rather than a verdict. Second, **panelist conformance:** it verifies that each panelist actually performed the task, produced a well-formed, on-task, non-degenerate review rather than an empty, off-topic, or malformed one, because a panelist that silently degrades reduces the effective N and, if undetected, inflates confidence in a thinner panel than the operator believes they have.

4.3 Recursive monitoring: n degrees

The consolidator is itself a model and can fail: it can drop a corroborated finding, promote an uncorroborated one, or mis-cluster. A meta-monitoring tier therefore applies the same corroboration-and-conformance discipline to the consolidator's output, and the construction recurses to n tiers. The design intent is that each tier catches the residual errors of the tier below. Section 6 shows the crucial qualification: this recursion multiplies the residual *only to the extent the tiers themselves are independent*, and because tiers are built from the same finite population of advanced base models, tier-to-tier independence is itself bounded, the same correlation problem, one level up.

5. Finding Schema and Matching

For corroboration to be meaningful, findings must be comparable. Each finding is a tuple of (location, class, severity, evidence). Two findings corroborate when their locations overlap and their classes match under a defined equivalence; the matching itself is a semantic judgment and is one more place a model can err, so match decisions are logged and are in scope for the conformance check. Throughout the error model we treat a 'finding' as an already-matched item, and fold matching error into the per-model rates.

6. A Correlation-Bounded Error Model

6.1 Definitions and the independent case

Consider a single candidate hallucinated finding, an assertion of a defect that does not exist. Let q be the per-model probability that a panelist independently reports this spurious item. Under the ≥k-of-N corroboration rule, the hallucination is promoted only if at least k of N panelists report it. If panelist errors are independent, the number reporting is X ~ Binomial(N, q), and the residual hallucination probability is

$$P\_FP = P(X \ge k) = \Sigma\_\{j=k\}\hat{}\{N\}\ C(N,j)\ q^j\ (1-q)\hat{}\{N-j\}.$$

Symmetrically, a real defect is missed when fewer than k of N panelists catch it; with per-model miss rate m and catch rate 1−m, the residual false-negative probability is P_FN = P(Binomial(N, 1−m) < k). With a majority-style threshold k = ⌈αN⌉ and α chosen above q, the independent residual decays exponentially in N by a Chernoff bound, this is the regime in which the 'add more reviewers' intuition is correct.

6.2 The correlated case: a shared-latent model

Real panelists do not err independently. We model the shared error with a latent item difficulty: for a given item, the propensity θ that a panelist reports it is drawn from a Beta(a,b) distribution with mean q and intra-class correlation ρ = 1/(a+b+1); conditional on θ, panelists report independently. The number reporting is then Beta-Binomial(N,a,b), and

$$P\_FP(\rho) = P(BetaBin(N,a,b) \geq k), \quad E[\theta] = q, \quad \rho = 1/(a+b+1).$$

The parameter ρ captures exactly the shared-data, shared-failure-mode coupling that model diversity is meant to reduce. As ρ→0 the beta-binomial collapses to the binomial and the independent case is recovered; as ρ grows, the tail thickens and corroboration loses power.

6.3 The correlation floor

The decisive result concerns the limit of large panels. By de Finetti's representation, as N→∞ with a fractional threshold α, the fraction of panelists reporting an item converges to θ, so

$$\lim\_\{N\to\infty\}\ P\_FP(\rho) = P(\theta \geq \alpha),$$

a quantity that depends only on the error distribution, not on N. Adding panelists cannot reduce residual hallucination below P(θ ≥ α); this is the **correlation floor.** Table 2 evaluates it. Because each monitoring tier is itself a consensus with its own floor, and tiers are built from the same base-model population, stacking tiers cannot escape a floor either, it can only reduce the above-floor component. The 'n degrees of monitoring drive error to zero' claim is therefore true in the ρ=0 idealization and false for any ρ>0.

6.4 Executed results

We fix q = 0.20, the per-model hallucination rate suggested by the largest package-hallucination study, which found 19.7% of model-recommended packages did not exist [6], and use the majority threshold k = ⌈0.6N⌉. Table 1 reports the residual hallucination probability across panel sizes and correlation levels; Table 2 reports the correlation floor; Figure 1 plots the same relationship. Table 3 reports the n-tier stacking behavior. A Monte-Carlo cross-check with 2×10^{6} trials at N=7, k=5, ρ=0.30 gives 8.235×10^{-2}, matching the closed-form 8.235×10^{-2} to within 0.00%, confirming the analytical computation.

Table 1. *Residual hallucination probability P(X ≥ k) versus panel size N and intra-panel error correlation ρ (per-model q = 0.20, threshold k = ⌈0.6N⌉).*

N (k)	ρ=0.00	ρ=0.05	ρ=0.15	ρ=0.30	ρ=0.50
3 (2)	1.0e-1	1.2e-1	1.4e-1	1.6e-1	1.8e-1
5 (3)	5.8e-2	7.9e-2	1.1e-1	1.5e-1	1.8e-1
7 (5)	4.7e-3	1.4e-2	4.1e-2	8.2e-2	1.3e-1
9 (6)	3.1e-3	1.3e-2	4.4e-2	9.0e-2	1.4e-1
15 (9)	7.8e-4	1.1e-2	4.8e-2	1.0e-1	1.5e-1
25 (15)	1.4e-5	3.9e-3	3.6e-2	9.1e-2	1.4e-1
51 (31)	2.3e-10	9.3e-4	2.5e-2	8.0e-2	1.4e-1

Reading: along the ρ=0 column, more reviewers help enormously (down to 2×10^{-10}). Along any ρ>0 column, the benefit of adding reviewers saturates quickly, past N≈7 there is almost no further reduction.

Table 2. *Correlation floor P(θ ≥ 0.6): the residual hallucination probability that no panel size and no number of monitoring tiers can beat, for q = 0.20.*

Intra-panel correlation ρ	Correlation floor
0.05	1.4e-4
0.15	2.0e-2
0.30	7.6e-2
0.50	1.3e-1

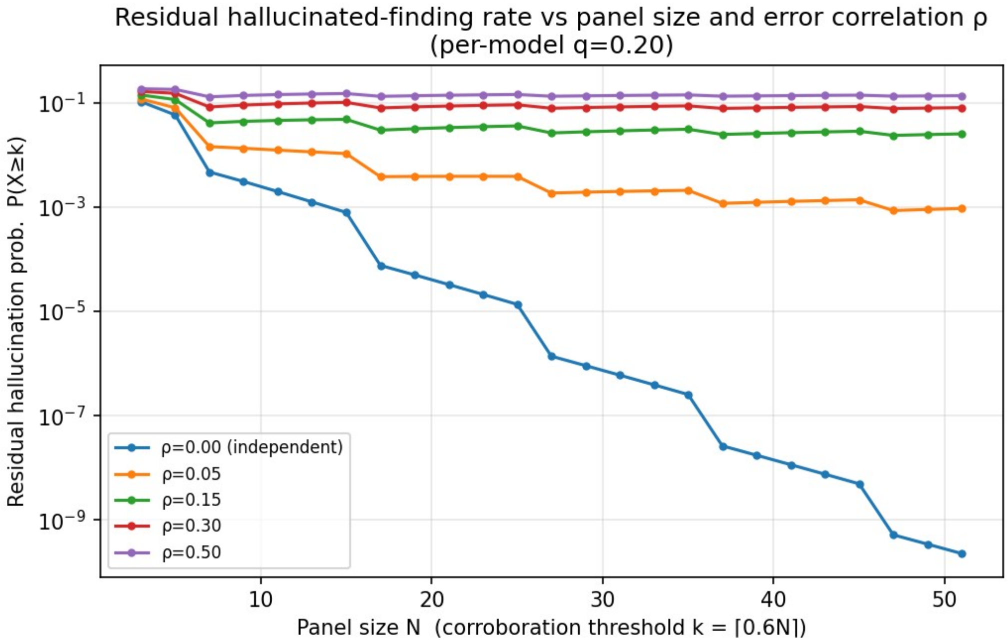

Figure 1. *Residual hallucinated-finding rate versus panel size N for several error-correlation levels (log scale). The ρ=0 curve falls without bound; every ρ>0 curve flattens onto its correlation floor.*

Table 3. *n-tier stacking, per-tier single-stage residual e = 1.1×10^{-1} (N=5, ρ=0.15). Independent tiers multiply; a 10% common-mode component across tiers floors the residual.*

Monitoring tiers n	Independent tiers (e^n)	Correlated tiers (c=0.10)
1	1.1e-1	1.1e-1
2	1.3e-2	2.3e-2
3	1.5e-3	1.3e-2
4	1.7e-4	1.2e-2

The uncomfortable finding. The three tables tell one story. Corroboration and recursive monitoring are genuinely powerful when errors are decorrelated, buying many orders of magnitude. But the residual is bounded below by a correlation floor that is invariant to panel size and tier count; at a plausible ρ=0.15 the floor is 2×10^{-2}, and no amount of additional reviewers or monitoring layers reduces it. The engineering leverage is therefore almost entirely in *lowering* ρ, maximizing genuine diversity among base models, rather than in scaling N or n, and ρ is exactly the quantity a simulator cannot honestly assume and a live study must measure.

7. Accountability and Regulatory Gates

A consensus reviewer that reduces hallucination does not by itself answer the question boards actually ask: when AI writes the code, who is responsible for it. The architecture must be embedded in an accountability and gating structure, or its statistical gains simply produce unaccountable merges faster.

The provenance problem. When a model authors a change, version-control blame points at whichever human committed it, often someone who does not fully understand what they landed. Responsibility does not transfer to the model. AI is a tool, not a liable party; the accountable human is whoever pressed merge, and the organization that shipped the artifact. The consensus layer changes how much assurance that human has before signing, but it does not move the locus of responsibility.

Provenance as a first-class artifact. An organization should be able to answer, for any system, how much of it was AI-authored or AI-assisted and who reviewed it. That requires tagging AI authorship in commit metadata and recording, for each change, which review panel and consolidation produced its sign-off. This is both an engineering-quality question and, increasingly, a regulatory one; the consensus layer is well placed to emit this record because it already produces structured, attributable review artifacts.

The inverted review bottleneck. Because machine authoring is cheap and review is the scarce resource, review capacity must scale with generation or unreviewed risk accumulates. The consensus layer is a way to scale review throughput, but it must scale review that is accountable and competent rather than review that rubber-stamps; a panel that agrees cheaply and wrongly is worse than a slow human, because it manufactures false assurance.

Regulatory gates as code. Compliance must be encoded in the pipeline, not asserted in a document. Gates should be automated and non-bypassable: license and provenance checks, SBOM generation, static and dynamic security analysis (SAST/DAST), policy-as-code admission control (e.g., OPA/Kyverno), build-provenance attestation (e.g., SLSA), and mandatory human sign-off on regulated changes [7,8]. The gate either passes or the build stops. The consensus layer feeds these gates, its findings become inputs to them, but it must never be permitted to satisfy or bypass a deterministic gate on their behalf; a probabilistic reviewer cannot discharge a categorical control.

Testing the gates. A control that is never tested is a control you do not have. The pipeline should be red-teamed with deliberately vulnerable or non-compliant code to confirm the gates catch it, and the gates should be audited on a schedule the way disaster-recovery is drilled. 'We have a gate' is not evidence; 'we tested the gate last Tuesday and it blocked X' is. The same discipline applies to the consensus layer itself: its false-negative rate on seeded, known defects is the only credible evidence that it is working, and it should be measured continuously rather than assumed from the architecture.

Where the human stays. The correlation floor of Section 6 has a direct governance consequence. For low-stakes, reversible changes, a residual of 10^{-2} with sampled human audit may be acceptable, and the human can move on-the-loop. For regulated or irreversible changes, a floor that cannot be driven arbitrarily low means human sign-off remains non-negotiable; the consensus layer raises the human's leverage and lowers their volume, but does not discharge their responsibility.

8. Vulnerability Ingestion and the “Vibe Coding” Attack Surface

The mechanism. When people ship code they did not write and do not understand, 'vibe coding', they ingest vulnerabilities wholesale. The model may reproduce insecure patterns from its training data, invent a dependency, or paste a configuration that quietly opens a door, and the person shipping it lacks the knowledge to notice. The consensus layer is partly a direct response to this: it inserts competent, corroborated scrutiny where the shipping human's own scrutiny is absent.

Hallucinated dependencies and slopsquatting. Code-generating models invent package names that do not exist. The largest study to date found that 19.7% of recommended packages were hallucinated across 2.23 million package references, and, critically for an attacker, 43% of hallucinated names recurred across repeated runs of the same prompt, making them predictable [6]. Attackers register these hallucinated names on public registries with malicious payloads, so the next developer who accepts the suggestion pulls in malware directly; this 'slopsquatting' vector is created entirely by model hallucination. The recurrence statistic is also why this failure is so dangerous for a consensus panel: a hallucination that a model makes reproducibly is precisely the kind of error that correlates across models trained on overlapping data, landing it in the high-ρ regime where corroboration is weakest.

Phishing and social engineering at scale. The same tools that write code write flawless, personalized phishing at volume, so a workforce is being targeted with better lures at the same time it is being trained to trust machine output, a dangerous combination that widens the human attack surface around the pipeline.

Defenses, and the consensus layer's role. The baseline defenses are enforced dependency allow-lists and private registries, mandatory scanning of all generated code regardless of author, secrets detection in the pipeline, and never letting AI-generated code skip the review path because a human 'kind of' looked at it, AI output is untrusted input until verified, with the same posture applied to an external contractor one has never met. The consensus layer contributes a specific check: a panelist that flags an imported dependency can have its existence verified deterministically against the registry, converting a probabilistic 'this package looks wrong' into a categorical gate result. But the architecture's honest limitation is exactly the shared false negative: a vulnerable pattern that every correlated panelist overlooks is the one case corroboration cannot catch, which is why the deterministic gates of Section 7 remain mandatory and why the consensus layer supplements rather than replaces them.

9. The Correlation-Measurement Study: A Live Experimental Protocol

The analytical model of Section 6 reduces the entire question of whether consensus review can be trusted to a single number: the inter-model error correlation. Everything else in the architecture, panel size, corroboration threshold, and number of monitoring tiers, is a design choice the operator controls, but the correlation is a property of the models themselves and cannot be chosen, assumed, or simulated honestly. This section specifies the study that measures it. It is the paper's central empirical contribution, and it is deliberately specified rather than executed here, for the reason given in Section 6: a simulator that assumes a correlation would merely return the assumption. We present the protocol at a level of detail sufficient for independent replication and pre-register its falsifiable predictions.

9.1 The estimand

Fix a corpus of code-review items, each carrying labeled ground truth about which findings are real. For an ordered pair of models (i, j) and an item, define the false-positive coincidence as the event that both models report the same spurious finding, and the false-negative coincidence as the event that both miss the same real defect. The quantities to be estimated are the pairwise error-correlation matrices *ρ_FP* and *ρ_FN*, together with the marginal per-model rates. We treat the shared-latent parameter of Section 6.2 as the target: fitting the beta-binomial (or, for pairwise structure, the tetrachoric) model to the observed coincidences recovers an estimate of ρ, and substituting the fitted parameters into $P(\theta \geq \alpha)$ yields the estimated correlation floor with a confidence interval. The two correlations are reported separately because they are not interchangeable in cost: ρ_FN governs shared blind spots, which ship defects, and is therefore the safety-critical quantity.

9.2 Corpus construction and ground truth

Ground truth is assembled from three complementary sources so that no single labeling weakness dominates. First, **seeded defects** are introduced into otherwise-correct code using published mutation operators and known vulnerable-pattern templates, giving exact, machine-checkable labels and controllable difficulty. Second, **historical real defects** are drawn from version-control history where a later fix commit localizes a genuine bug, giving realistic, non-synthetic defects at the cost of noisier labels. Third, **expert-annotated findings** on real diffs are labeled independently by at least two qualified reviewers, blind to the models' outputs, with a documented adjudication procedure and a reported inter-annotator agreement (Cohen's kappa) so that label reliability is itself measured rather than assumed. A dedicated stratum of **hallucinated-dependency items** is included, in which the corpus contains diffs importing plausible but non-existent packages, because the package-hallucination literature reports that models reproduce the same fabricated names across runs [6] and this stratum is where ρ is predicted to be highest.

9.3 Panel configuration

The panel comprises N advanced models chosen to span providers and base-model families rather than to span prompts to one model, because the estimand is precisely the correlation that shared pre-training induces. Each model receives the same diff and task under an independent context with no visibility of any other model's output, and prompt and sampling diversity are applied within a logged, reproducible configuration. To separate genuine model-diversity effects from prompt-diversity effects, the design crosses *model family* with *prompt variant* so that the marginal contribution of adding a same-family model versus a cross-family model can be estimated directly. Every finding is emitted in the structured schema of Section 5, and cross-model finding matching is performed by a defined equivalence whose decisions are audited on a human-labeled subsample so that matching error is bounded and reported rather than hidden inside the correlation estimate.

9.4 Measurement procedure and estimators

For each item the panel produces per-model findings; each finding is scored against ground truth as a true positive, false positive, true negative, or false negative after matching. From the resulting per-model event vectors the study computes the pairwise coincidence tables, estimates ρ_{FP} and ρ_{FN} by the tetrachoric correlation and by an intraclass-correlation fit, and fits the hierarchical beta-binomial to recover the population ρ used in Section 6. Consolidation is then evaluated at a range of corroboration thresholds k by replaying the recorded panel outputs through the consolidator, so that the consolidated false-positive and false-negative rates, the consolidator's own conformance-detection rate, and the marginal value of each additional monitoring tier are all measured on the same runs. All estimates are reported with bootstrap confidence intervals, clustered by item to respect the nested structure. A power analysis fixes the corpus size: to estimate a pairwise correlation to a half-width of roughly 0.05 at 95% confidence requires on the order of a few thousand labeled items per stratum, and the protocol sizes each stratum accordingly rather than reporting whatever precision a convenience sample happens to yield.

A measurement pitfall the estimator must correct. The false-negative correlation is estimated over an unbiased universe, because every ground-truth finding is enumerated whether or not any model reports it. The false-positive correlation is not so lucky: the only spurious findings an operator observes are those at least one model actually reported, so estimating hallucination correlation over reported findings alone conditions on the joint event that someone fired and biases the estimate toward zero, and in practice toward spurious anti-correlation. The protocol therefore estimates false-positive correlation over an item-level universe in which every reviewed item is a unit and the event is whether a model committed any hallucination on it, so that items on which the whole panel stayed silent remain in the denominator. On a synthetic corpus with a known injected correlation the item-level estimator recovers it while the reported-findings estimator collapses to near zero, which is why the released harness computes and reports both and treats the item-level figure as primary.

Table 4. *Live correlation-measurement study at a glance.*

Element	Specification
Ground-truth sources	Seeded (mutation/vuln templates), historical fix-localized defects, dual-expert

	annotation
Panel	N cross-provider, cross-family models; independent contexts; family x prompt crossed design
Primary estimands	Pairwise ρ_FP and ρ_FN; population ρ; implied floor P(θ≥α) with CI
Estimators	Tetrachoric correlation, intraclass correlation, hierarchical beta-binomial fit
Uncertainty	Item-clustered bootstrap CIs; reported label kappa; audited matching error
Sizing	Power-analysed: ~few thousand labeled items/stratum for ρ half-width ≈0.05
Released	Corpus, per-model outputs, labels, matching decisions, analysis code

9.5 Pre-registered, falsifiable predictions

The model of Section 6 makes specific predictions that the study can refute. Registering them before data collection converts the study from a demonstration into a test. Table 5 lists the primary predictions and the observation that would falsify each.

Table 5. *Pre-registered predictions and their falsification conditions.*

Prediction	Falsified if
P1. ρ_FN > ρ_FP: shared blind spots exceed shared hallucinations	Measured ρ_FN ≤ ρ_FP across strata
P2. Hallucinated-dependency stratum shows the highest ρ_FP	That stratum's ρ_FP is not the largest
P3. A same-family addition raises ρ more than a cross-family addition	Family has no measurable effect on ρ
P4. Implied floor exceeds 1e-2 for N up to 15 at realistic ρ	Floor falls below 1e-2 while N≤15
P5. Beyond a small N, added panelists give <2x residual reduction	Residual keeps falling ~geometrically in N

Prediction P4 is the one with governance teeth: if the measured floor sits above 10^{-2} for practical panel sizes, then autonomous review cannot meet the residual most organizations would demand for regulated changes, and the human-on-the-loop boundary of Section 7 is not a temporary limitation but a structural one. If instead the corpus reveals genuinely low correlation among a reachable set of diverse models, the same protocol quantifies exactly how much autonomy is justified and for which change classes.

9.6 Cost, ablations, and open release

Because a panel of N models across n tiers costs on the order of N times n model invocations per item, the study records per-item token and dollar cost alongside accuracy, so that the marginal reliability purchased by each additional model or tier can be reported against its marginal cost. Planned ablations remove one model family at a time to measure its decorrelating contribution, vary the corroboration threshold to trace the false-positive against false-negative trade-off, and disable the consolidator conformance check to measure how often a silently degraded panelist would otherwise inflate apparent agreement. The corpus, the full per-model outputs, the labels and adjudication records, the matching decisions, and the analysis code are released so that the correlation estimates are independently recomputable, which matters especially here because the entire argument of the paper turns on a quantity that must be measured in the open rather than asserted. The released ingestion harness accepts exactly the output a live N-model panel produces, one JSON record per item carrying the ground-truth findings and each model's reported findings, so that a pilot on a few hundred labeled diffs plugs directly into the estimators above and returns the pairwise correlations, the population fit, the implied floor with confidence intervals, and the consolidation sweep, with no simulation in the path.

9.7 Why simulation cannot substitute

The study is not replaceable by a larger or more elaborate simulation, and the distinction is not a matter of scale. A simulation must encode a correlation structure as an input, and its output is then a consequence of that input; using it to estimate the real correlation would be circular. Only live, heterogeneous models scored against independent ground truth produce an estimate of ρ that was not placed there by the experimenter. This is the same principle by which the prior AORE work declined to report cross-model discrimination from a scripted harness, and it is why this protocol, rather than any additional analytical result, is the paper's central contribution and its designated next step.

10. Evaluation of the Error Model

10.1 Scope and honesty disclosure

Every quantitative result in this paper comes from an executed analytical and Monte-Carlo computation over the error model of Section 6, implemented in a released script, not from live model panels. The computation demonstrates the mathematics of consensus and monitoring, how residual error behaves as a function of panel size, corroboration threshold, error correlation, and monitoring tiers, and its Monte-Carlo cross-check confirms the closed forms to within 0.00%. It does not demonstrate anything about the error rates or, especially, the error correlation of real, diverse models on real diffs, and this paper does not claim otherwise.

10.2 Executed analytical and Monte-Carlo evaluation

Tables 1–3 and Figure 1 are the executed results: the residual-hallucination surface over (N, ρ), the correlation floor, the n-tier stacking behavior, and the Monte-Carlo agreement check. Their evidentiary value is precise but bounded, they establish that the architecture's benefit is governed by error correlation and that a floor exists, using q anchored to a published hallucination rate [6] and ρ swept across a plausible range. They do not establish where on that range real panels sit; that is the object of the study in Section 9.

10.3 Agent-level simulation cross-check

A second, released simulation reproduces the closed-form results under an independent parameterization and adds pipeline detail the closed form omits. It models each panelist's per-item decision through a shared-latent Gaussian-copula so that the pairwise correlation is set directly, adds an explicit consolidator error rate, and stacks monitoring tiers with a common-mode coupling. Its beta-binomial validation matches the closed form of Section 6 to within one percent across the checked configurations. The added detail surfaces one operational finding not visible in the closed form: once the panel residual falls below the consolidator's own error rate, the consolidator becomes the dominant floor, a concrete instance of the single-point-of-failure noted in Section 4.3. This strengthens rather than weakens the paper's thesis, since it identifies a second floor, the consolidator's own reliability, stacked beneath the correlation floor.

10.4 A genuine negative result

The model produced a finding its designers did not want: past a small panel size, adding reviewers or monitoring tiers yields almost nothing once errors are correlated, because the residual is pinned to the correlation floor (Table 2). The engineering implication is uncomfortable for the 'more monitoring is more safety' intuition, the dominant lever is decorrelating the panel, not enlarging it, and the consolidator itself remains a single point of failure whose own error is bounded by the same wall. This is reported as a limitation of the architecture, not reframed as a strength.

11. Discussion

Adoption path. The architecture supports incremental adoption: begin with the panel and consolidator running in shadow behind existing human review, measure the consolidated review's false-negative rate against human findings on seeded and real defects, and only widen autonomy, moving humans from in-the-loop to on-the-loop with sampled audit, for change classes where the measured residual is acceptable and the blast radius is bounded. Regulated and irreversible changes retain human sign-off regardless of measured residual.

Cost and latency economics. A panel of N models across n tiers costs on the order of N×n model invocations per review, so the scheme is expensive, and the floor result sharpens the economics: because benefit saturates quickly in N once $\rho>0$, the cost-effective configuration is a small, maximally-diverse panel with few tiers rather than a large homogeneous one. Spending the marginal dollar on a genuinely different base model buys more than spending it on another copy of a model already in the panel.

Relation to human review. The correct framing is on-the-loop, not out-of-the-loop. Consensus review raises the assurance a human sign-off carries and lowers the volume a human must personally inspect, but for the changes that matter most it does not remove the human, because the correlation floor denies the arbitrarily-low residual that full autonomy would require.

12. Threats to Validity

Construct validity. The architecture assumes agreement tracks correctness. If panelists share a bias, agreement tracks the shared bias instead, promoting a corroborated but wrong finding or suppressing a corroborated blind spot; this is the same failure the correlation floor formalizes, and it means high agreement is not, on its own, evidence of correctness.

Internal validity. All results derive from a single author's analytical model and simulation; the independence and correlation structure is imposed by the beta-binomial assumption rather than measured, and the true error-generating process across real models may not be beta-binomial.

External validity. The premise of the whole design is model diversity, and market consolidation onto a few base models and shared pre-training corpora directly erodes it, pushing real ρ upward and the floor higher over time, so the architecture may become weaker precisely as the ecosystem it runs in matures.

The correlation-measurement problem. The one number that decides practical value, ρ, is also the hardest to estimate and cannot be assumed honestly; until it is measured on live heterogeneous panels, the floor's location, and thus whether autonomous review is safe for a given change class, remains unknown.

13. Conclusion

Consensus-layer review with recursive monitoring is a real and quantifiable reduction in hallucinated review outcomes, and under decorrelated errors the reduction is large, many orders of magnitude in panel size. It is not, however, an elimination. The residual is bounded below by a correlation floor set by how much the panelists' errors coincide, and that floor is invariant to the number of reviewers and the number of monitoring tiers; the claim that n degrees of monitoring drive hallucination to zero holds only in an idealization that models sharing data and failure modes violate. The practical consequence is that the engineering effort belongs in decorrelating the panel rather than enlarging it, that the human moves on-the-loop but not out of it for changes that matter, and that consensus review feeds the deterministic accountability and supply-chain gates rather than replacing them. The empirical claims here are analytical and simulated, and the one question the model cannot answer about itself, where real, diverse models sit on the correlation axis, requires live multi-model measurement, which we specify and leave as the necessary next step.

Declaration of Generative-AI Use

During the preparation of this work the author used a generative-AI assistant for literature grounding, for implementing and running the analytical and Monte-Carlo error-model computation, and for drafting and language editing. The author reviewed and edited all content and takes full responsibility for the paper, including the correctness of the derivations and the honesty of the demarcation between analytical and live-evaluation claims.

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